Muhammad Farhan Fahad

AI Engineer

+923499485775

furhikhan@gmail.com

E11, Islamabad

https://github.com/frhaan05

https://www.linkedin.com/in/frhaan05/

I'm AI Engineer with a strong interest in Generative AI and intelligent systems. Experienced in building AI applications such as chatbots and AI-powered solutions using Python and modern AI frameworks. Possess a solid understanding of LLM, RAG, vector databases, Agentic AI, AI Agents and prompt engineering. Also have foundational knowledge of Machine Learning, Deep Learning, and Computer Vision. Passionate about developing innovative AI solutions and applying advanced AI technologies to solve real-world problems.

EDUCATION

Bachelor of Science in Artificial Intelligence
Air University Islamabad, Pakistan
Sep 2021 – July 2025

High Secondary School (Pre-Engineering)
Bradford Science college
March 2019 – July 2021

KEY PROJECTS

Personal ChatBot

Developed an interactive AI chatbot capable of text interaction, speech processing, and image generation.

- Developed using **FastAPI**, **Groq, LLM API**, and **Replicate API**.
- Built a **custom frontend** with HTML, CSS, and JavaScript.
- Implemented **rule-based and AI-driven responses** for dynamic conversations.
- Integrated **speech-to-text (STT)** and **text-to-speech (TTS)** functionalities.
- Enabled image generation functionality using AI APIs.
- GitHub Link: https://github.com/frhaan05/Personal-ChatBot

Technologies: Python, FastAPI, AI APIs, HTLM, CSS, JavaScript

Early Edge Programming (FYP)

Designed an educational platform to teach programming concepts through interactive games and AI-assisted learning.

- Developed an interactive learning environment for students to practice coding concepts
- Implemented a feedback system and performance tracking dashboard
- Integrated AI-driven assistance to guide learners during programming tasks
- GitHub Link: https://github.com/frhaan05/KidsCodingGame

Technologies: HTML, CSS, JavaScript, MySQL, PHP

Auto-Suggestion Model using NLP

Built a natural language processing model to predict the next word in a sentence.

- Implemented an N-gram language model for word prediction
- Trained and evaluated the model using text datasets
- Improved prediction accuracy through preprocessing and probability optimization

Technologies: Python, NLTK, NLP techniques

Lane Line Detection

- Built a computer vision system for lane detection using Canny Edge Detection, Hough Transform, and ROI masking.
- Enhanced accuracy with color filtering (HSV/HLS) and Gaussian smoothing for robust performance under varying conditions.
- GitHub Link: https://github.com/frhaan05/LaneLineDetection

Face Emotions Project

Developed a deep learning model to recognize human emotions from facial images.

- Trained convolutional neural networks to classify facial expressions
- Processed image datasets and optimized model performance
- Implemented real-time emotion detection using computer vision
- GitHub Link: https://github.com/frhaan05/Face_Emotion_Detection

Technologies: Python, TensorFlow, OpenCV

TECHNICAL SKILLS

Programming Languages: Proficient in Python, C++ and JavaScript
Machine Learning & Deep Learning: TensorFlow, PyTorch, Scikit-learn
Natural Language Processing: NLTK, Text Processing, Language Modeling
Computer Vision: OpenCv, Image Processing
Data Analysis: Pandas, NumPy, Matplotlib
Web Development: Flask, FastAPI, HTML, CSS, JavaScript, Node.js
Database: SQL, Pinecone, Chroma
Generative AI Concepts: Prompt Engineering, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Vector Databases, AI Agents

CERTIFICATIONS

- LLM Engineering: Master AI, Large Language Models & Agents *(certificate)*
- Deep Learning with Keras and Tensorflow *(certificate)*
- Pytorch ultimate team from basic to cutting edge *(certificate)*
- Advanced PyTorch Techniques and Applications *(certificate)*
- Building and Training Neural Networks with PyTorch
- Foundations and Core Concepts of PyTorch
- Operating System Foundation

STRENGTHS

- Strong theoretical foundation in Artificial Intelligence
- Problem-solving and analytical thinking
- Ability to design and build AI systems
- Continuous learner of emerging AI technologies